

MAT 549 HW04

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Problem 1 (12-8). Let M be a smooth n -manifold, and let A be a smooth covariant k -tensor field on M . If $(U, (x_i))$ and $(\tilde{U}, \tilde{x}_j)$ are overlapping smooth charts on M , we can write

$$A = \sum_{1 \leq i_1, \dots, i_k \leq n} A_{i_1, \dots, i_k} dx_{i_1} \otimes \cdots \otimes dx_{i_k} = \sum_{1 \leq j_1, \dots, j_k \leq n} \tilde{A}_{j_1, \dots, j_k} d\tilde{x}_{j_1} \otimes \cdots \otimes d\tilde{x}_{j_k}.$$

Compute a transformation law analogous to (11.7) expressing the component functions $A_{i_1, \dots, i_k}$ in terms of $\tilde{A}_{j_1, \dots, j_k}$.

Solution. By equation (11.7) in Lee,

$$dx_i = \sum_{j=1}^n \frac{\partial \tilde{x}_j}{\partial x_i} d\tilde{x}_j.$$

By definition of tensors, the smooth function commutes with $\otimes$, so

$$\sum_{1 \leq i_1, \dots, i_k \leq n} A_{i_1, \dots, i_k} dx_{i_1} \otimes \cdots \otimes dx_{i_k} = \sum_{1 \leq i_1, \dots, i_k \leq n} \tilde{A}_{i_1, \dots, i_k} \left(\sum_{j_1=1}^n \frac{\partial \tilde{x}_{j_1}}{\partial x_{i_1}} d\tilde{x}_{j_1} \right) \otimes \cdots \otimes \left(\sum_{j_k=1}^n \frac{\partial \tilde{x}_{j_k}}{\partial x_{i_k}} d\tilde{x}_{j_k} \right).$$

This means that the coefficient of a particular simple tensor $d\tilde{x}_{\ell_1} \otimes \cdots \otimes d\tilde{x}_{\ell_k}$ is

$$A_{\ell_1, \dots, \ell_k} = \boxed{\sum_{1 \leq i_1, \dots, i_k \leq n} \tilde{A}_{i_1, \dots, i_k} \frac{\partial \tilde{x}_{\ell_1}}{\partial x_{i_1}} \cdots \frac{\partial \tilde{x}_{\ell_k}}{\partial x_{i_k}}}. \quad \square$$

Problem 2 (14-1). Show that covectors $\omega_1, \dots, \omega_k$ on a finite-dimensional vector space V are linearly dependent if and only if $\omega_1 \wedge \cdots \wedge \omega_k = 0$.

Solution. If $\omega_1, \dots, \omega_k$ are linearly dependent, there exist real numbers $c_1, \dots, c_k$, not all zero, such that $c_1\omega_1 + \cdots + c_k\omega_k = 0$. Since reordering the vectors does not change whether their wedge product is nonzero (only affecting the sign), we can assume that $c_1 \neq 0$, so that $\omega_1 = \frac{c_2}{c_1}\omega_2 \wedge \cdots \wedge \frac{c_k}{c_1}\omega_k$. Then

$$\omega_1 \wedge \omega_2 \wedge \cdots \wedge \omega_k = \sum_{j=2}^k \frac{c_j}{c_1} \omega_j \wedge \omega_2 \cdots \omega_k = 0$$

where each summand is zero because ω_j appears twice in the wedge.

Conversely suppose that $\omega_1, \dots, \omega_k$ are linearly independent, so that we can find covectors $\omega_{k+1}, \dots, \omega_n$ to make the list $\omega_1, \dots, \omega_n$ a basis of V . Then $\omega_1 \wedge \cdots \wedge \omega_n$ is nonzero (since otherwise $\Lambda^n V^*$ would be zero-dimensional, because any element of $\Lambda^n V^*$ can be written as a scalar multiple of $\omega_1 \wedge \cdots \wedge \omega_n$). This implies $\omega_1 \wedge \cdots \wedge \omega_k$ is nonzero too, because the wedging with zero always yields zero. $\square$

Problem 3 (14-5). Let M be a smooth n -manifold with or without boundary and let $(\omega_1, \dots, \omega_k)$ be an ordered k -tuple of smooth 1-forms on an open subset $U \subseteq M$ such that $(\omega_1|_p, \dots, \omega_k|_p)$ is linearly independent for each $p \in U$. Given smooth 1-forms $\alpha_1, \dots, \alpha_k$ on U such that

$$\sum_{i=1}^k \alpha_i \wedge \omega_i = 0,$$

show that each α_i can be written as a linear combination of the ω_i with smooth coefficients.

Solution. Let $p \in M$ be arbitrary, and let U be a small enough open set around p such that $(\omega_1, \dots, \omega_k)$ can be extended to a list $(\omega_1, \dots, \omega_n)$ for which $(\omega_1|_p, \dots, \omega_n|_p)$ is a basis for T_p^*M for each $p \in U$. Then, for each $1 \leq i \leq k$ and $1 \leq j \leq n$ there exist $A_{i,j} \in C^\infty(U)$ such that

$$\alpha_i = \sum_{j=1}^n A_{i,j} \omega_j.$$

For a particular $1 \leq i \leq k$,

$$0 = \left(\sum_{i'=1}^n \alpha_{i'} \wedge \omega_{i'} \right) \wedge (\omega_1 \wedge \dots \wedge \omega_{i-1} \wedge \omega_{i+1} \wedge \dots \wedge \omega_k) = (-1)^{i-1} \alpha_i \wedge (\omega_1 \wedge \dots \wedge \omega_k)$$

This implies that $(\alpha_i, \omega_1, \dots, \omega_k)$ are linearly dependent at each $p \in M$, so, in U , the coefficients $A_{\ell,j}$ of ω_ℓ for $k+1 \leq \ell \leq n$ in the coordinate representation of α_i are all zero (noting that the ω_j are nonzero because they span T_p^*M at each $p \in U$). Thus $\alpha_i = \sum_{j=1}^k A_{i,j} \omega_j$ in U .

If $\alpha_i = \sum_{j=1}^k A_{i,j} \omega_j$ in U and $\alpha_i = \sum_{j=1}^k B_{i,j} \omega_j$ in V for some open subsets $U, V \subseteq M$, then by the linear independence of the $\omega_j|_p$ we can conclude that $A_{i,j} = B_{i,j}$ on $U \cap V$. Thus the coefficients $A_{i,j}^{U_p}$ in a neighborhood U_p of $p \in M$ agree with each other inside intersections of sets in the open cover $\{U_p\}$, so they define smooth functions $A_{i,j} : M \rightarrow \mathbb{R}$ globally. $\square$

Problem 4 (14-6). Define a 2-form ω on $\mathbb{R}^3$ by

$$\omega = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy.$$

- Compute ω in spherical coordinates (ρ, ψ, θ) defined by $(x, y, z) = (\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi)$.
- Compute $d\omega$ in both Cartesian and spherical coordinates and verify that both expressions represent the same 3-form.
- Compute the pullback $\iota_{S^2}^* \omega$ to S^2 , using coordinates (φ, θ) on the open subset where these coordinates are defined.
- Show that $\iota_{S^2}^* \omega$ is nowhere zero.

Solution. (a) We have

$$\begin{aligned} dx &= \sin \varphi \cos \theta d\rho + \rho \cos \varphi \cos \theta d\varphi - \rho \sin \varphi \sin \theta d\theta \\ dy &= \sin \varphi \sin \theta d\rho + \rho \cos \varphi \sin \theta d\varphi + \rho \sin \varphi \cos \theta d\theta \\ dz &= \cos \varphi d\rho - \rho \sin \varphi d\varphi \\ dx \wedge dy &= \rho^2 \cos \varphi \sin \varphi d\varphi \wedge d\theta + \rho \sin^2 \varphi d\rho \wedge d\theta \\ dy \wedge dz &= -\rho \sin \theta d\rho \wedge d\varphi + \rho^2 \sin^2 \varphi \cos \theta d\varphi \wedge d\theta - \rho \sin \varphi \cos \varphi \cos \theta d\rho \wedge d\theta \\ dz \wedge dx &= \rho \cos \theta d\rho \wedge d\varphi + \rho^2 \sin^2 \varphi \sin \theta d\varphi \wedge d\theta - \rho \sin \varphi \cos \varphi \sin \theta d\rho \wedge d\theta \end{aligned}$$

so

$$\omega = \rho^3 \sin \varphi d\varphi \wedge d\theta.$$

- In Cartesian coordinates,

$$d\omega = dx \wedge dy \wedge dz + dy \wedge dz \wedge dx + dz \wedge dx \wedge dy = 3dx \wedge dy \wedge dz,$$

noting that the cyclic permutations of x, y , and z are even. In spherical coordinates, we get

$$d\omega = 3\rho^2 \sin \varphi d\rho \wedge d\varphi \wedge d\theta$$

which is equal to the expression in Cartesian coordinates by Corollary 14.21 in Lee:

$$d\rho \wedge d\varphi \wedge d\theta = \det \begin{pmatrix} \sin \theta \cos \theta & \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \\ \cos \varphi & -\rho \sin \varphi & 0 \end{pmatrix} dx \wedge dy \wedge dz = \rho^2 \sin \varphi d\rho \wedge d\varphi \wedge d\theta.$$

(c) In spherical coordinates, $\iota_{S^2}(\varphi, \theta) = (1, \varphi, \theta)$, so the pullback of ω by the inclusion is

$$\iota_{S^2}^* \omega = \sin \varphi d(\varphi \circ \iota_{S^2}) \wedge d(\theta \circ \iota_{S^2}) = \sin \varphi d\varphi \wedge d\theta.$$

(d) The coordinate system (φ, θ) is defined away from the poles, and in this region $0 < \varphi < \pi$, so $\omega = \sin \varphi d\varphi \wedge d\theta$ is nonzero away from the poles. A similar argument applied after swapping the x and z axes shows that ω is nonzero away from the intersections of S^2 with the x -axis, so $\iota_{S^2}^* \omega$ is nowhere zero. $\square$